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MySQL - SQL Injection

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SQL Injection

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SQL injection is a code injection technique that can destroy your database. SQL injections are a common web hacking technique.

SQL injections are when attackers insert malicious SQL code into user-input fields, and this way can read, modify, or delete sensitive data in a database.

SQL injections usually occur when you ask a user for input, like username/userid, and instead of giving a name/id, the attacker inserts an SQL command that executes something in your database.

Look at the following example which creates a `SELECT` statement by adding a variable (txtUserId) to a select string. The variable is fetched from user input (getRequestString):

Example

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The rest of this chapter describes the potential dangers of using user input in SQL statements.

The next chapter show the most effective methods to **prevent SQL injections**, by using Prepared Statements.

SQL Injection Based on 1=1 is Always True

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Look at the example above again. The the SQL code was to select a user with a given user id.

If there is nothing to prevent a user fr input, the user can enter some "smart" input like this:

UserId: 105 OR 1=1

Then, the SQL statement will look like this:

```
SELECT * FROM Users WHERE UserId = 105 OR 1=1;
```

The SQL above is valid and will return ALL rows from the "Users" table, since **OR 1=1** is always TRUE.

Does the example above look dangerous? What if the "Users" table contains names and passwords?

A hacker might get access to all the user names and passwords in a database, by simply inserting 105 OR 1=1 into the input field.

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SQL Injection Based on OR ""="" is Always True

Here is an example of a user login on a web site:

Username:

John Doe

Password:

myPass

Example

```
uName = getRequestString("username");
uPass = getRequestString("userpassword");

sql = 'SELECT * FROM Users WHERE Name ="' + uName + '" AND Pass ="' + uPass +
'"'
```

Result

```
SELECT * FROM Users WHERE Name ="John Doe" AND Pass ="myPass"
```

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" or ""="

Password:

" or ""="

The SQL statement will now look like this:

Result

```
SELECT * FROM Users WHERE Name ="" or ""="" AND Pass ="" or ""=""
```

The SQL above is valid and will return ALL rows from the "Users" table, since **OR ""=""** is always TRUE.

SQL Injection From Batched SQL Statements

Batched SQL statements is a group of two or more SQL statements, separated by semicolons.

The SQL statement below will return all rows from the "Users" table, then delete the "Suppliers" table.

Example

```
SELECT * FROM Users; DROP TABLE Suppliers;
```

Look at the following example:

Example

And the following input:

User id: 105; DROP TABLE Suppliers

The valid SQL statement would look like this:

Result

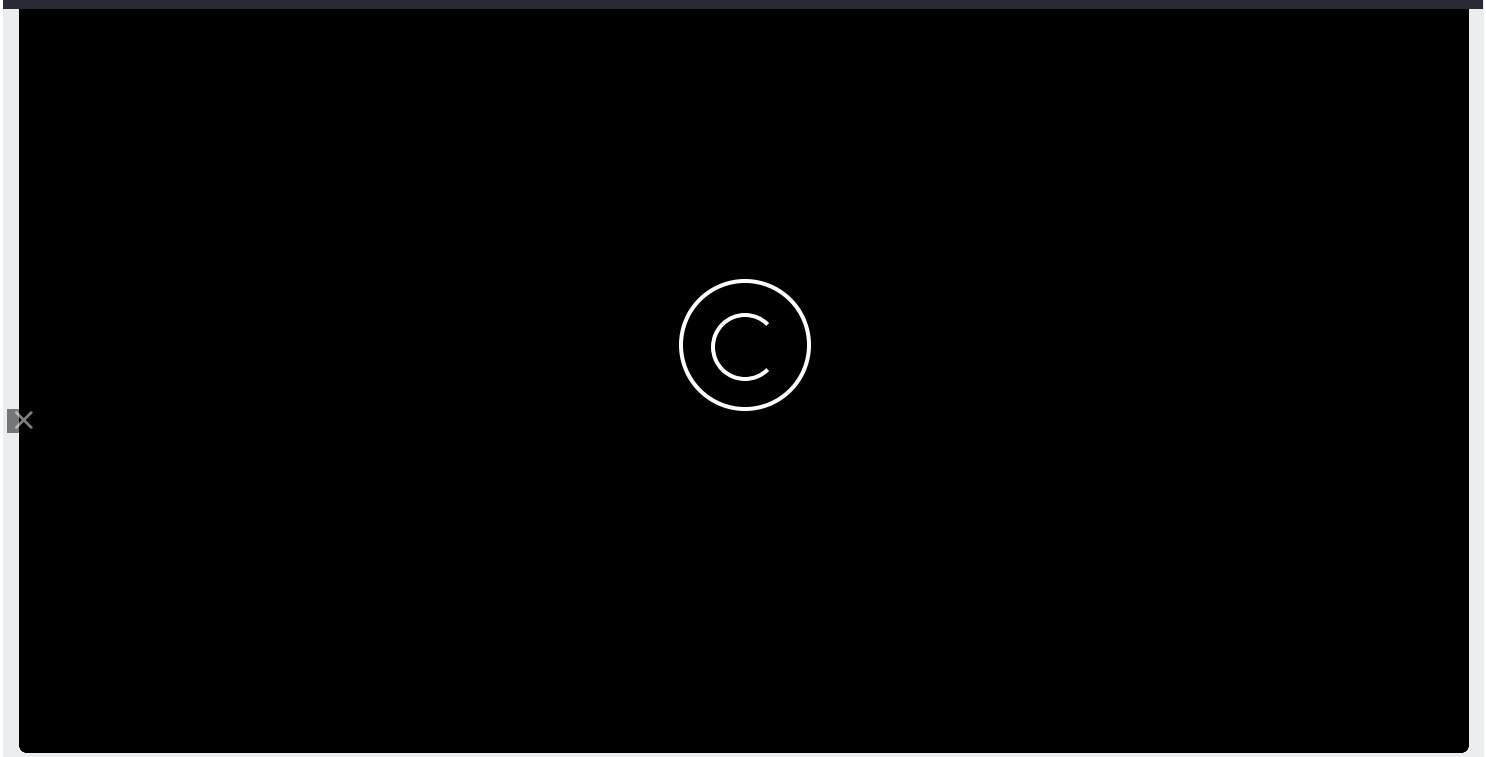
```
SELECT * FROM Users WHERE UserId = 105; DROP TABLE Suppliers;
```

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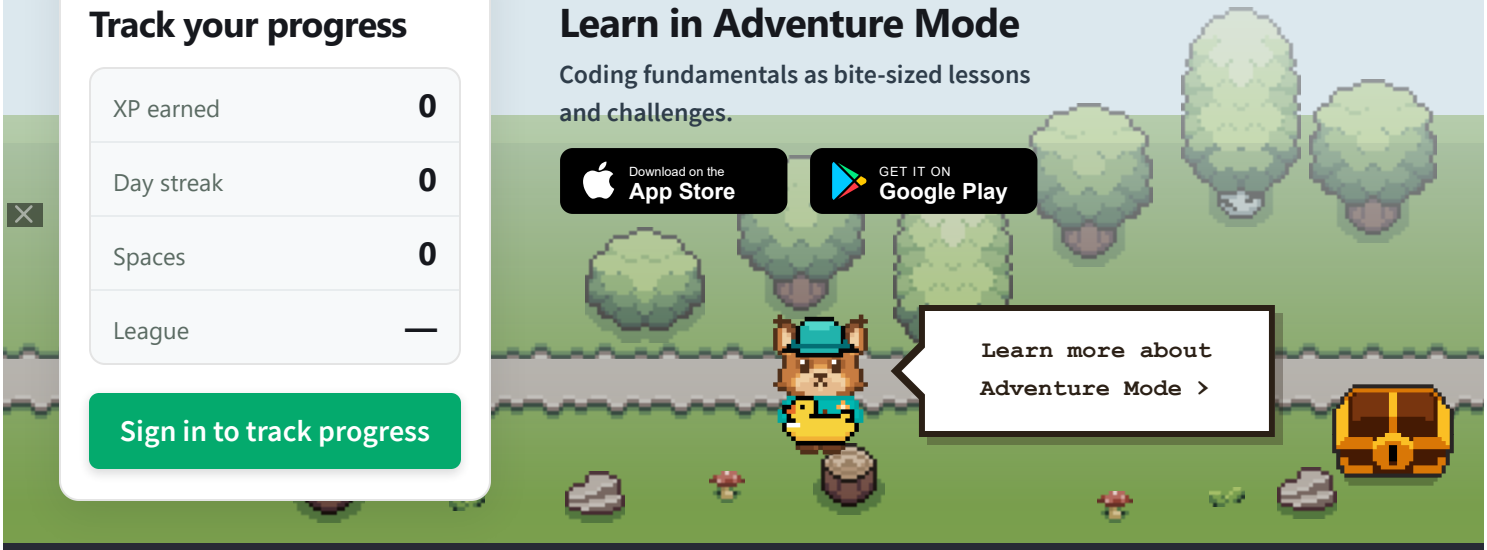
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